

Rifah Jahan Tamanna

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OBJECTIVE

Motivated and detail-oriented Full Stack Developer with a strong foundation in modern web technologies, database management, and problem-solving. Passionate about building fast, scalable, and user-friendly web applications while continuously learning new technologies and best practices. Seeking an opportunity to contribute to a collaborative development team, grow as a software engineer, and create impactful digital solutions.

EDUCATION

BRAC University

Dhaka, Bangladesh

Bachelor of Science in Computer Science

May 2026

- **CGPA:** 3.6 / 4.0
- **Relevant Coursework:** Machine Learning, Neural Networks, Database Systems, Data Structures & Algorithms, Software Engineering, Operating Systems

SKILLS

Frontend: React.js, Next.js, Tailwind CSS, JavaScript

Backend & Database: Node.js, Express.js, PostgreSQL, Supabase, REST APIs

AI & LLM Integration: Gemini API, Claude Code, ChatGPT

Data & Analytics: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, Power BI

Tools: Git, GitHub, VS Code, Vercel, Railway, Postman

Soft Skills: Leadership, cross-functional collaboration, analytical thinking, technical documentation, fast learning, adaptability

PROJECTS

VocabNex — AI-Powered Vocabulary Builder [🐙 GitHub](#) | [🔗 Live Demo](#)

- Designed and built a full-stack AI-powered web application from scratch using Next.js, React, Supabase, and the Gemini API.
- Implemented a smart definition caching layer: AI-generated definitions are stored in a shared Supabase database so repeat words load instantly with zero API calls, reducing repeated API calls and improving response times for previously searched words.
- Built active recall features including animated flashcards with progress tracking and an AI-generated multiple choice quiz mode, with scores saved per user for progress monitoring.
- Used AI-driven development tools (Claude Code) throughout the build process, demonstrating proficiency in modern AI-assisted workflows.
- **Tech:** Next.js 16, React.js, Tailwind CSS, Supabase, Gemini API, Vercel

DevPulse — Internal Issue & Feature Tracker [🐙 GitHub](#) | [🔗 Live Demo](#)

- Built a RESTful backend for bug and feature request tracking with JWT authentication, bcrypt password hashing, and role-based access control (contributor vs. maintainer)
- Designed a modular Express.js + TypeScript architecture with centralized error handling, auth middleware, and environment-based configuration; deployed live on Railway with PostgreSQL
- **Tech:** Node.js, Express.js, PostgreSQL, JWT Authentication, GitHub Actions, Railway

Data Analysis Dashboard (Power BI) [🔗 Drive Folder](#)

- Designed interactive dashboards from raw Excel and SQL datasets, performing end-to-end data cleaning, transformation, and structured visual reporting to surface actionable business insights.
- Created summary views and trend analyses enabling faster, evidence-based decision-making.

- **Tech:** Power BI, Microsoft Excel, SQL, Power Query

BRISC2025 Brain Tumor Classification & Segmentation [GitHub](#)

- Implemented and compared U-Net and Attention U-Net architectures for simultaneous MRI tumor segmentation and multi-class classification (glioma, meningioma, pituitary, no tumor)
- Designed a dual-task loss combining Dice + BCE (segmentation) and Cross-Entropy (classification) with joint optimization
- **Tech:** Python, PyTorch, NumPy, Pandas, Matplotlib

Multimodal Music Genre Classification using VAE [GitHub](#)

- Built a multimodal pipeline combining audio (MFCC) and lyrics (TF-IDF) features using a Beta-VAE and Neural Autoencoder to learn a joint latent representation for music genre classification
- Applied K-Means clustering on fused latent vectors; evaluated against baselines (raw features, PCA, AE) using Silhouette Score, NMI, ARI, and Cluster Purity
- **Tech:** Python, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

EXTRACURRICULAR & LEADERSHIP

Aspire Leadership Program (Harvard-affiliated)

Participant

Ongoing

- Competitively selected from a global applicant pool for a programme focused on ethical leadership, strategic thinking, and cross-cultural communication.
- Collaborating with international peers on structured challenges, strengthening adaptability, initiative, and the ability to work across diverse teams.

BRAC University Computer Club

Executive Member

2022 – 2024

- Co-organised tech events, workshops, and seminars for 100+ students; managed documentation, logistics, reporting, and member coordination across the executive team.
- Developed event planning, stakeholder communication, and team management skills in a fast-paced, student-run environment.

BRAC University Robotics Club

Apprentice

2022 – 2024

- Contributed to team-based technical builds and problem-solving challenges, strengthening collaborative execution and logical thinking under pressure.

Duke of Edinburgh's Award

Bronze Level Achiever

- Completed across volunteering, physical, and skill development sections — demonstrating personal accountability, consistency, and commitment to self-improvement.